The documentation and process conversion measures necessary to comply with this document shall be completed by 16 October 2014.

INCH-POUND

MIL-PRF-19500/773
16 July 2014

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DUAL TRANSISTOR, UNITIZED, NPN/PNP, SILICON, LOW-POWER, TYPES M19500/773-01, JAN, JANTX, JANTXV, JANS, JANSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for unitized, dual transistors which contain a pair of electrically isolated NPN and PNP silicon, low-power transistors in one package. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (8 lead flatpack).

1.3 Maximum ratings unless otherwise specified T_A = +25°C.

Types	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_C	$I_{CM(1)}$	I_B	$I_{BM(1)}$
	V dc	V dc	V dc	mA dc	mA dc	A dc	mA dc	mA dc
NPN	60	60	6	800	800	4	200	400
PNP	-60	-60	-6	-800	-800	-4	-200	-400

Types	P_T T_A= +25°C		P_T T_C= +25°C		$R_{\theta JA}$		$R_{\theta JC}$		T_J and T_{STG}
	One section	Both sections	One section	Both sections	One section	Both sections	One section	Both sections	
	W	W	W	W	°C/W	°C/W	°C/W	°C/W	°C
NPN	0.8	1.4	5	7	180	125	35	25	-65 to +200
PNP	0.8		5		180		35		

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1.3 Maximum ratings unless otherwise specified T_A = +25°C. - Continued.

NOTES:
(1) Collector peak current (tp<5ms)
(2) Base peak current (tp<5ms).
(3) See 3.3 for abbreviations.

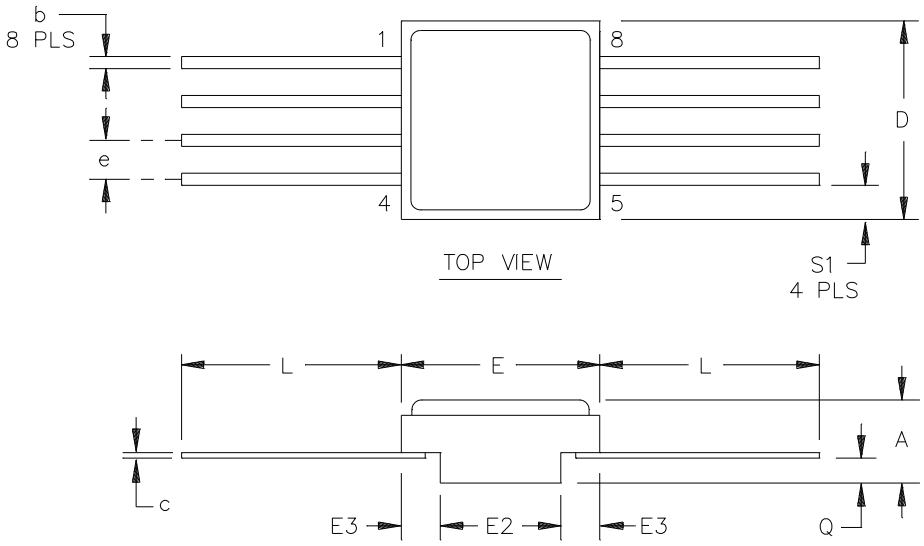
1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C (1)

Limits	I_{CBO}	I_{EBO}	$V_{BE(on)}$	$V_{CE(sat)}$ (2)		h_{FE} (2)	
	V_{CB}=60V	V_{EB}=6V	V_{CE}=2V I_C=100mA	I_C=0.8A I_B=40mA	Ic=2A I_B=100mA	I_C=100mA V_{CE}=2V	I_C=1A V_{CE}=2V
	nA	nA	mV	mV	mV		
Min			600			80	160
Max	100	100	720	160 (NPN) 180 (PNP)	380 (NPN) 440 (PNP)		400

	$t_{(on)}$ V_{CC} = 10 V dc I_C = 0.8 A dc Ibon=80 mA	$t_{(off)}$ V_{CC} = 10 V dc IC = 0.8 A dc Iboff=-80 mA
	ns	ns
Max NPN	175	2500
Max PNP	150	1000

(1) For PNP type, voltage and current are negative
(2) Pulsed (see 4.5.1).

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Ref.	DRAWING inches			DRAWING mm			NOTES
	MIN	TYP	MAX	MIN	TYP	MAX	
A	.088	.096	.104	2.24	2.44	2.64	
b	.015	.017	.019	0.38	0.43	0.48	
c	.004	.005	.006	0.10	0.13	0.16	
D	.250	.255	.260	6.35	6.48	6.61	
E	.250	.255	.260	6.35	6.48	6.61	
E2	.170	.175	.180	4.32	4.45	4.58	
E3	.035	.040	.045	0.88	1.01	1.14	
e		.050			1.27		
L	.256	-	.291	6.51	-	7.38	
Q	.026	.031	.036	0.66	0.79	0.92	
S1	.036	.044	.052	0.92	1.12	1.32	
N	08			08			

NOTES:
1. Dimensions are in inches.
2. Millimeters equivalents are given for general information only.

FIGURE 1. Physical dimensions for device type M19500/773-01 (FLAT-8).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$R_{\theta JA}$	Thermal resistance junction to ambient.
$R_{\theta JC}$	Thermal resistance junction to case.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Marking. Devices shall be marked in accordance with MIL-PRF-19500.

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3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.4, 1.5, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Dual transistor unitized, NPN/PNP, silicon, low-power devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening.

4.3.1 Screening of encapsulated devices (JANS, JANTX, JANTXV levels). Screening of packaged devices shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement	
	JANS	JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.3).	Thermal impedance (see 4.3.3).
9	I_{CBO1}, h_{FE2}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO1}, h_{FE2} ΔI_{CBO1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value or 30mV , whichever is greater	I_{CBO1}, h_{FE2}
12	See 4.3.2	See 4.3.2
13	ΔI_{CBO1}= 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value or 30mV, whichever is greater, subgroup 2 and 3 of table I herein.	ΔI_{CBO1}= 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value or 30mV, whichever is greater, subgroup 2 of table I herein.

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (V_C and V_H where appropriate). See group E, subgroup 4 herein.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of subgroup 1 and 2, of table I herein, inspection only (table E-VIb, group B, subgroup 1 is not required to be performed since solderability and resistance to solvents testing is performed in table I herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. End-point electrical measurements shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and E-VIB (JANTX & JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection (JANS), table E-VIa of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table E-VIa, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve a T_J = +225°C minimum.
B6	3131	$R_{\theta JA}$ only.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = 150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours T_A = +200°C. n = 22, c = 0.

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.3.1 Group C inspection (JANS), table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, 3 ounce weight: Three bends of 15 degrees for M19500/773-01 (22 devices, 2 leads each device).
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3 and 4.3.2).
C6	1026	1,000 hours at V_{CB} = 10 V dc; power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3 n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Group C sample selection. Samples selected from group C inspection shall be chosen at random from any lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a group using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS type shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750 for total ionizing dose, or method 1017 of MIL-STD-750 for neutron fluence, as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 3/						
Visual and mechanical examination 4/	2071					
Solderability 4/ 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 4/ 5/ 6/	1022	n = 15 devices, c = 0				
Temp cycling 4/ 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 5/		Table I, subgroup 2				
Bond strength 4/ 5/	2037	Precondition T_A = +250°C at t = 24 hours or T_A = +300°C at t = 2 hours, n = 11 wires, c = 0				
Decap internal visual (design verification) 5/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance	3131	See 4.3.3	$Z_{\theta JX}$			°C/W
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 60V dc pulsed (see 4.5.1).	I_{CBO1}		100	nA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 6 V dc pulsed (see 4.5.1).	I_{EBO}		100	nA dc
Breakdown voltage, collector-base	3001	Bias condition D; I_C = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)CBO}$	60		V dc
Breakdown voltage, collector-emitter	3011	Bias condition D; I_C = 1 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Breakdown voltage emitter-base	3026	Bias condition D, I_C = 10 µA dc, pulsed (see 4.5.1)	$V_{(BR)EBO}$	6		V dc
Collector-emitter saturated voltage	3071	I_C = 0.8 A dc, I_B = 40 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$			
NPN					160	mV dc
PNP					180	mV dc
Collector-emitter saturated voltage	3071	I_C = 2 A dc, I_B = 100 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$			
NPN					380	mV dc
PNP					440	mV dc
Base-emitter voltage (non saturated)	3066	Condition B, V_{CE} = 2V;I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{BE(ON)}$	600	720	mV dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 100 mA dc, pulsed (see 4.5.1)	h_{FE1}	100		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE2}	160	400	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 60 V dc	I_{CBO2}		10	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 100 mA dc, pulsed (see 4.5.1)	h_{FE1}	40		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, f = 1 MHz	C_{obo}			
NPN					25	pF
PNP					60	pF
Saturated turn-on time		I_c=0.8A, V_{CC}=10V, $V_{BE(off)}$=-5V, $I_{B(on)}$=-$I_{B(off)}$= 80mA (see figure 4)	t_{on} (NPN)		175	ns
Saturated turn off time			t_{off} (NPN)		2,500	ns
Saturated turn-on time		I_c=2A, V_{CC}=10V, $V_{BE(off)}$=-5V, $I_{B(on)}$=-$I_{B(off)}$= 200mA (see figure 5)	t_{on} (PNP)		150	ns
Saturated turn off time			t_{off} (PNP)		1,000	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan see MIL-PRF-19500.
2/ For PNP transistor voltage and current are negative.
3/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
4/ Separate samples may be used.
5/ Not required for JANS devices.
6/ Not required for laser marked devices.
7/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.

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TABLE II. Group D inspection and end-point limits.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0 V				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc; pulsed (see 4.5.1)	I_{CBO}		200	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6.0 V dc; pulsed (see 4.5.1)	I_{EBO}		200	nA dc
Breakdown voltage, collector - base	3001	Bias condition D; I_C = 100 µA dc	$V_{(BR)CBO}$	60		V dc
Breakdown voltage, collector - emitter	3011	Bias condition D; I_C = 1 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc
Breakdown voltage Emitter-base	3026	Bias condition D, I_C = 10 µA dc, pulsed (see 4.5.1)	$V_{(BR)EBO}$	6		V dc
Collector – emitter saturated voltage	3071	I_C = 0.8 A dc; I_B = 40 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$			
NPN					184	mV dc
PNP					207	mV dc
Collector – emitter saturated voltage	3071	I_C = 2 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$			
NPN					437	mV dc
PNP					506	mV dc
Base-emitter voltage (non saturated)	3066	Condition B, V_{CE} = 2V;I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{BE(ON)}$	600	828	mV dc
Forward-current transfer ratio	3076	V_{CE} = 2V dc, I_C = 100 mA dc, pulsed (see 4.5.1)	[h_{FE1}] 5/	[50]		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	[h_{FE2}] 5/	[80]	400	

See footnotes at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Steady-state total dose irradiation	1019	Gamma exposure V_{CES} = 48 V				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc; pulsed (see 4.5.1)	I_{CBO}		200	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6.0 V dc; pulsed (see 4.5.1)	I_{EBO}		200	nA dc
Breakdown voltage, collector - base	3001	Bias condition D; I_C = 100 µA dc	$V_{(BR)CBO}$	60		V dc
Breakdown voltage, collector – emitter	3011	Bias condition D; I_C = 1 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc
Breakdown voltage Emitter base	3026	Bias condition D, Ic = 10 µA dc, Pulsed (see 4.5.1)	$V_{(bre)EBO}$	6		V dc
Collector – emitter saturated voltage	3071	I_C = 0.8 A dc; I_B = 40 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$			
NPN					184	mV dc
PNP					207	mV dc
Collector-emitter Saturated voltage	3071	Ic = 2A dc, IB = 100 ma dc, Pulsed (see 4.5.1)	$V_{CE(sat)2}$			
NPN					437	mV dc
PNP					506	mV dc
Base-emitter voltage (non saturated)	3066	Condition B, V_{CE} = 2V; IC = 100 mA dc; pulsed (see 4.5.1)	$V_{BE(ON)}$	600	828	mV dc
Forward-current transfer ratio	3076	V_{CE} = 2V dc, I_C = 100 mA dc, pulsed (see 4.5.1)	$[h_{FE1}]$ 5/	[50]		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	$[h_{FE2}]$ 5/	[80]	400	

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding UB suffix version unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019 of MIL-STD-750, for how to determine $[h_{FE}]$ by first calculating the delta $(1/h_{FE})$ from the pre-and post-radiation h_{FE}. Notice the $[h_{FE}]$ is not the same as h_{FE} and cannot be measured directly. The $[h_{FE}]$ value can never exceed the pre-radiation minimum h_{FE} that it is based upon.
6/ For PNP transistor voltage and current are negative

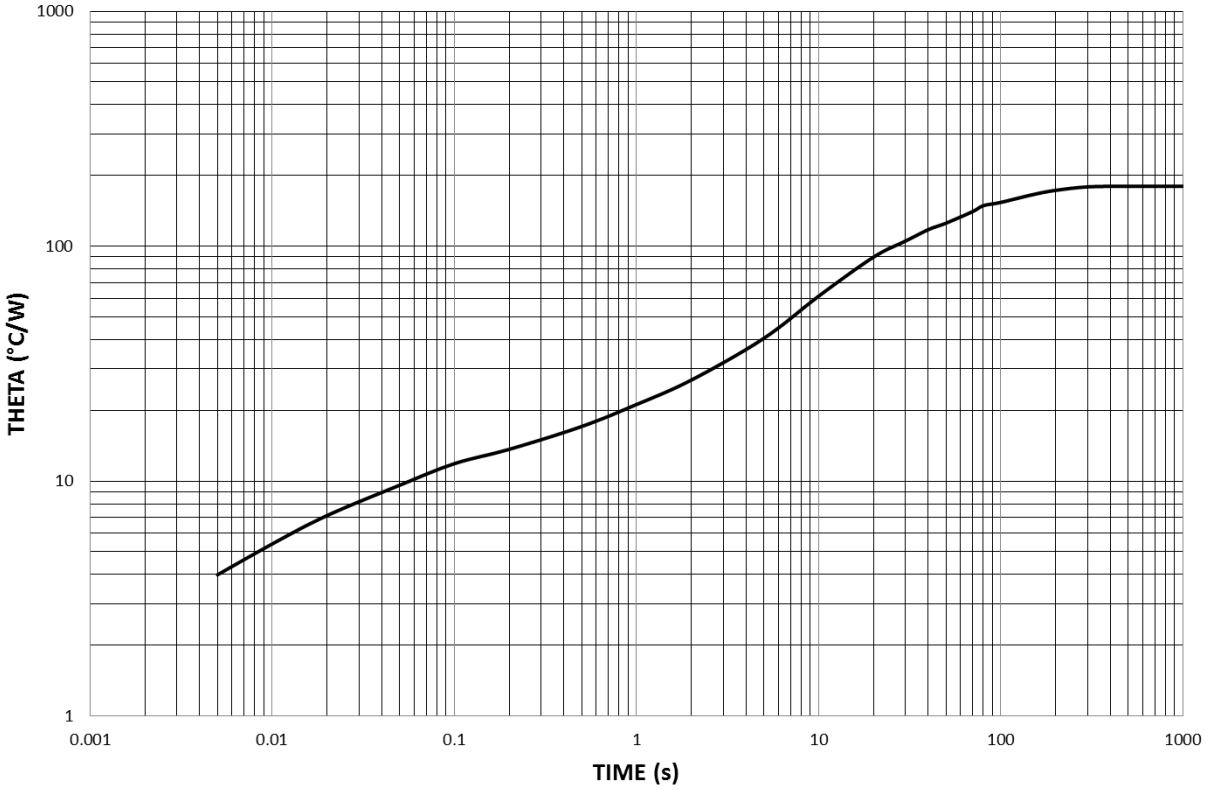
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TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices, c = 0
Intermittent life 1/	1037	Intermittent operation life: V_{CB} = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal impedance curves.		See MIL-PRF-19500.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 6			11 devices
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

1/ For PNP voltage and current are negative.

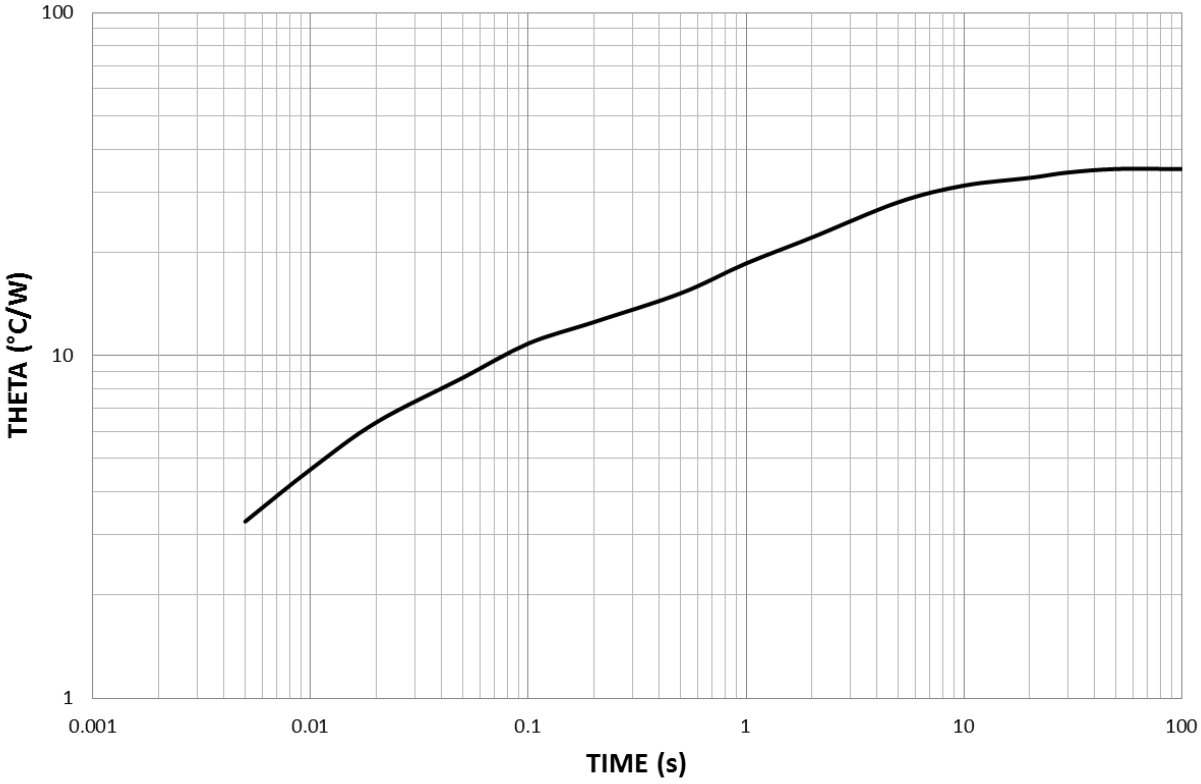
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Thermal impedance M19500/773-01, FP8 package (Socket mount)
Thermal resistance =180°C/W one side, 125 °C/W both sides operating in parallel

FIGURE 2. Thermal impedance graph ($R_{\theta JA}$) for M19500/773-01 (FP8) NPN PNP one section

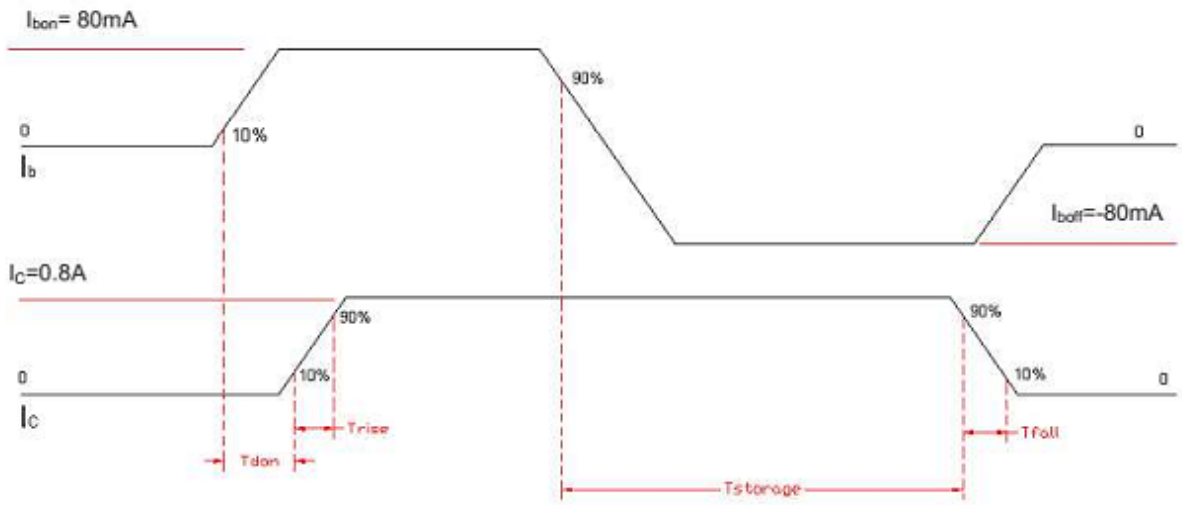
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Thermal impedance M19500/773-01, FP8 package (Heat sink mount)
Thermal resistance =35°C/W one side, 25 °C/W both sides operating in parallel

FIGURE 3. Thermal impedance graph ($R_{\theta JC}$) for M19500/773-01 (FP8) one section-

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$t_{on} = t_{don} + t_{rise}$

$t_{off} = t_{storage} + t_{fall}$

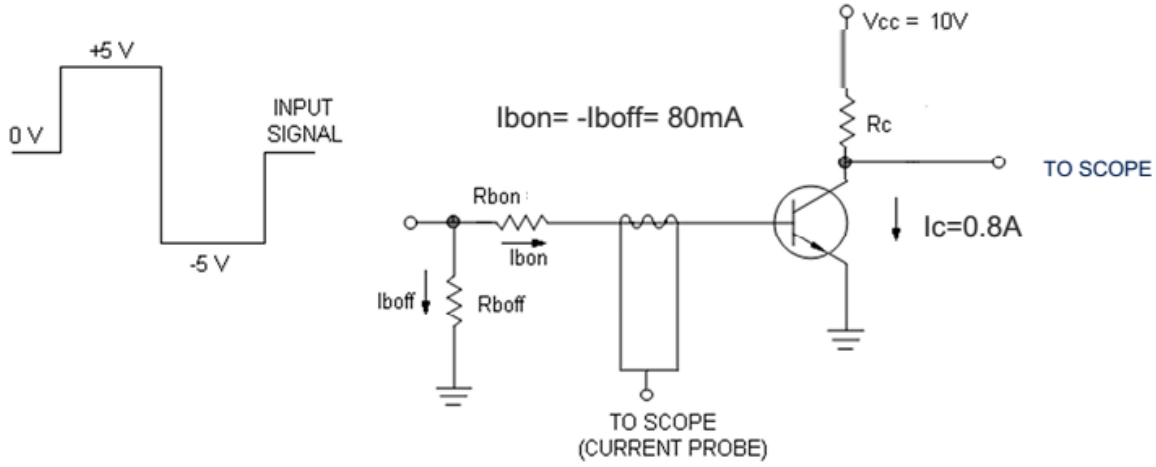
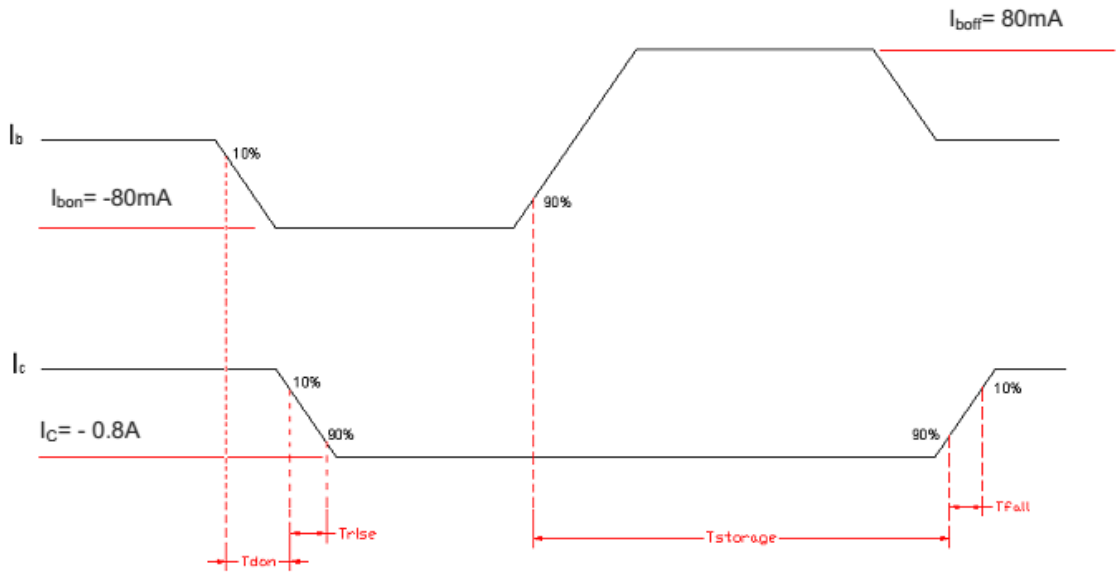


FIGURE 4. Both sides: Saturated turn-on/turn off time test circuit-NPN

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$t_{on} = t_{don} + t_{rise}$ $t_{off} = t_{storage} + t_{fall}$

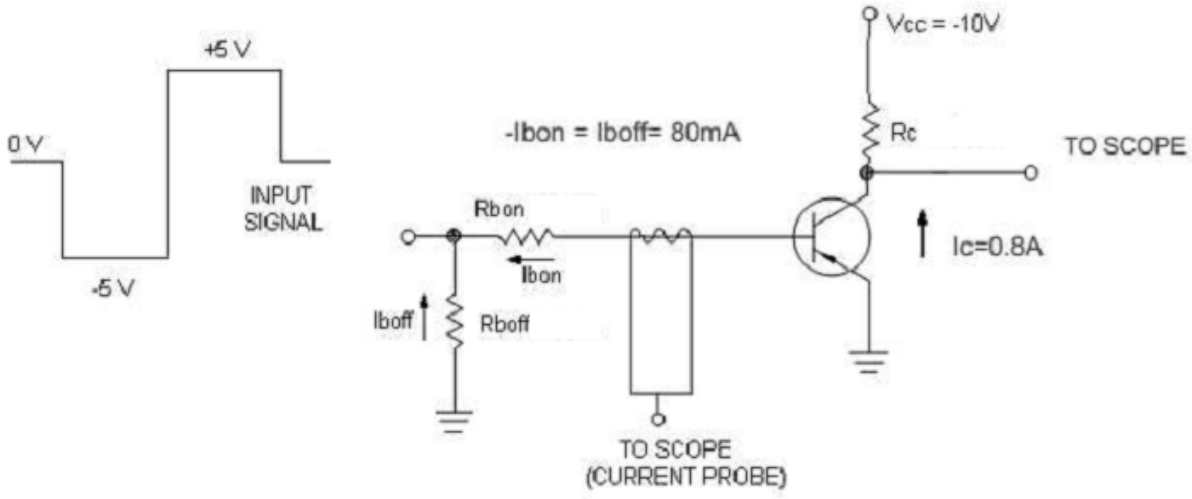


FIGURE 5. Saturated turn-on/turn off time test circuit-PNP

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see title and section 1.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it must be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

(Project 5961-2014-073)

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